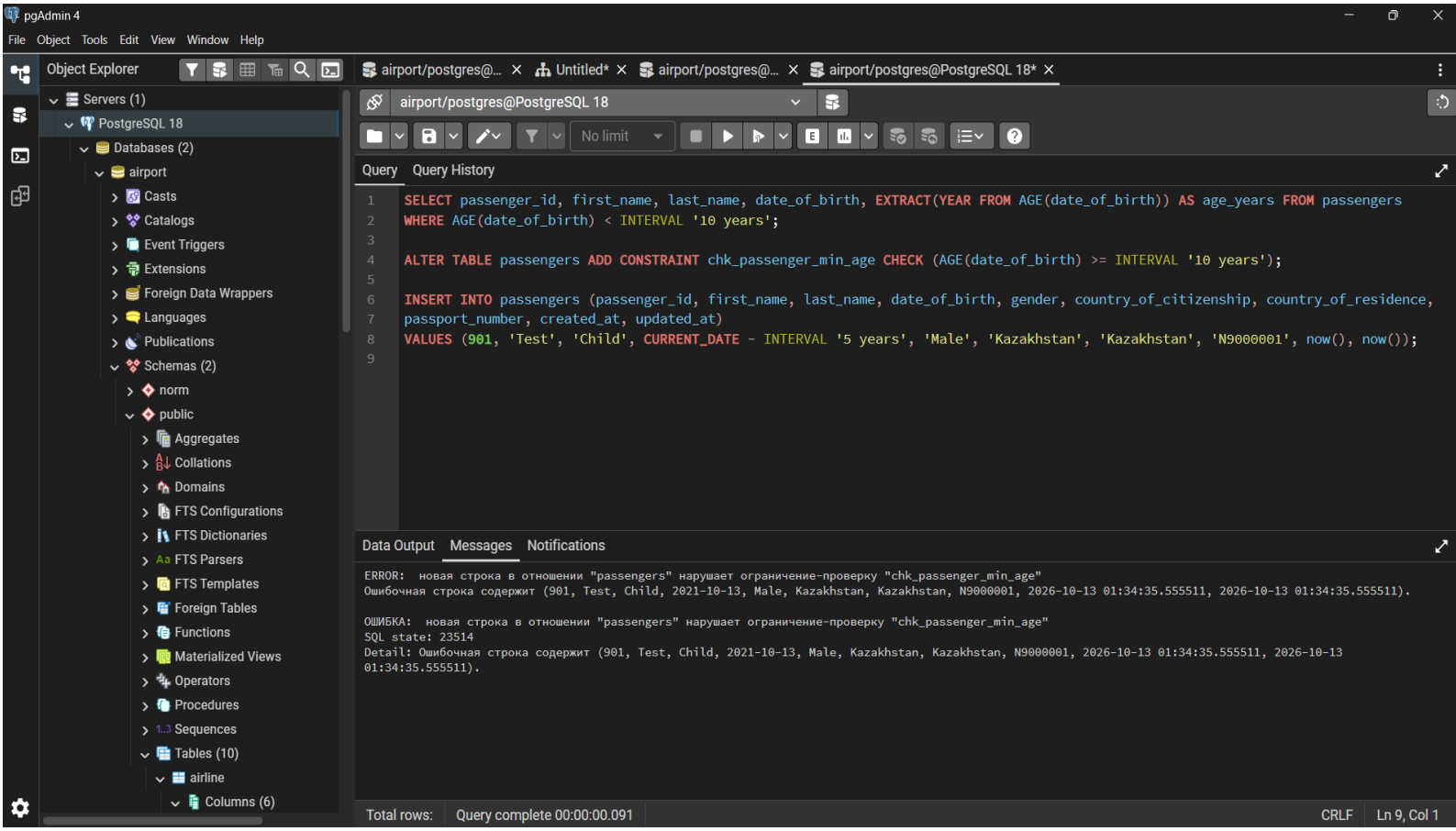
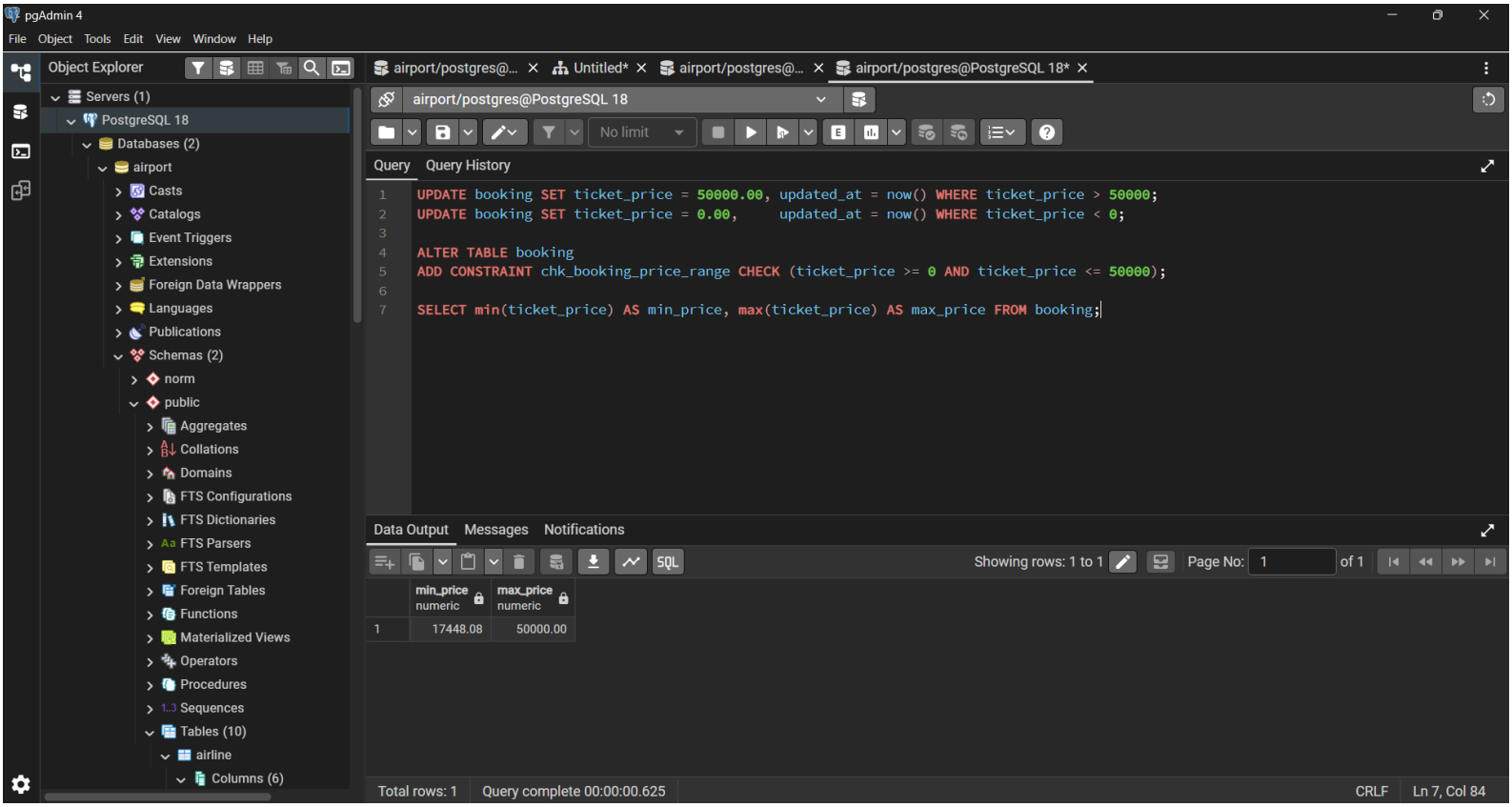
Laboratory work 5

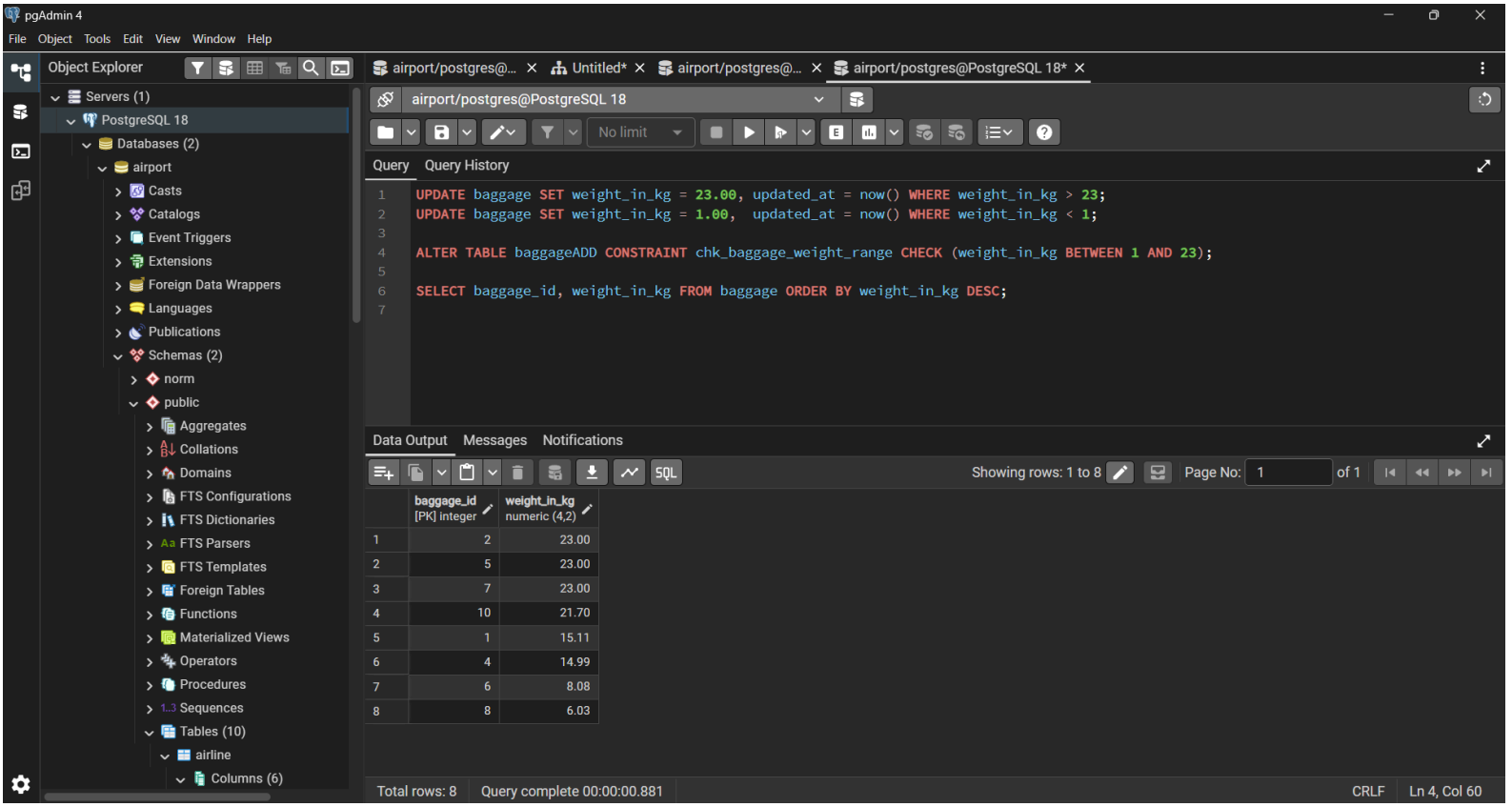
Task 1



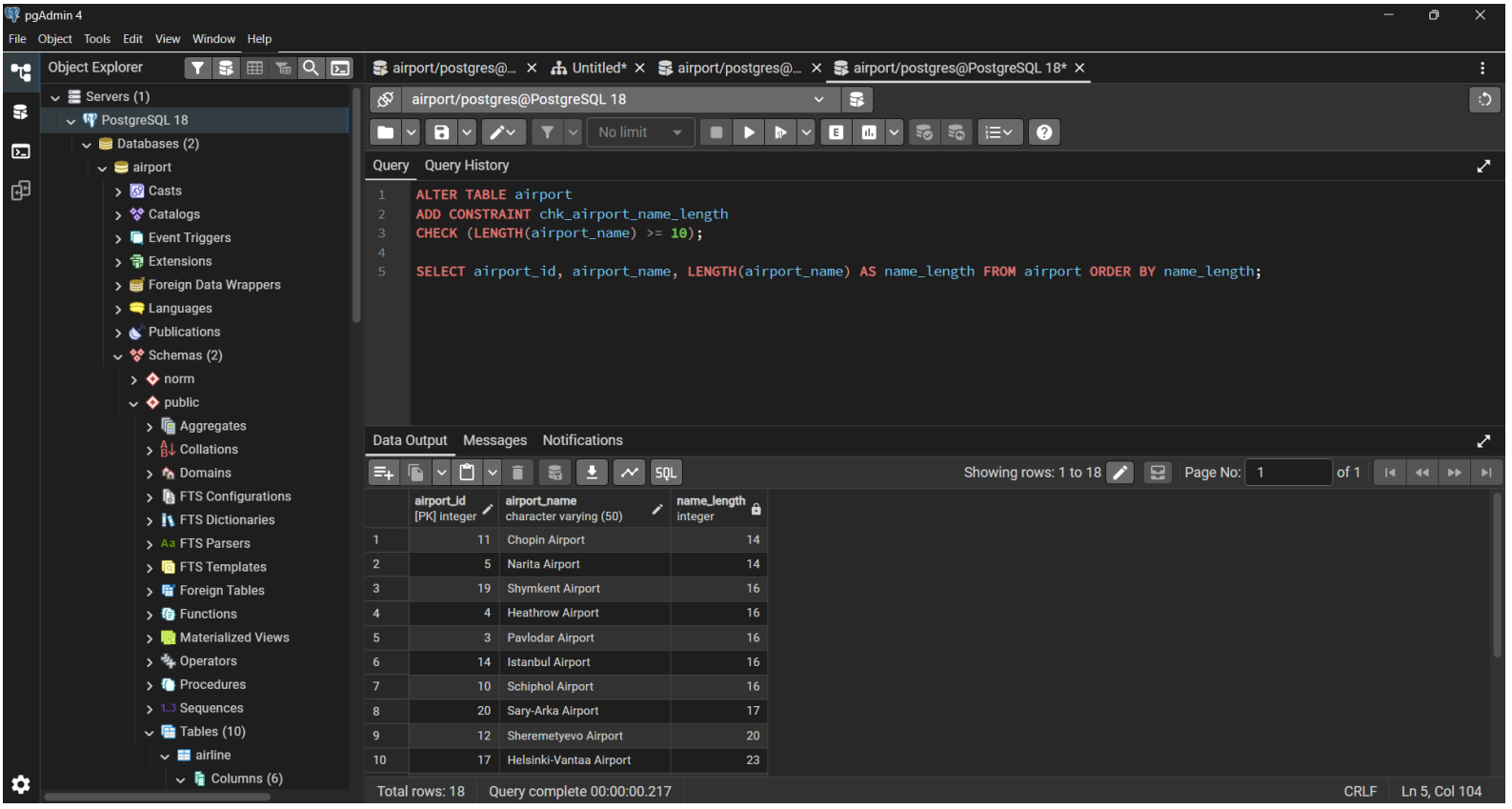
Task 2



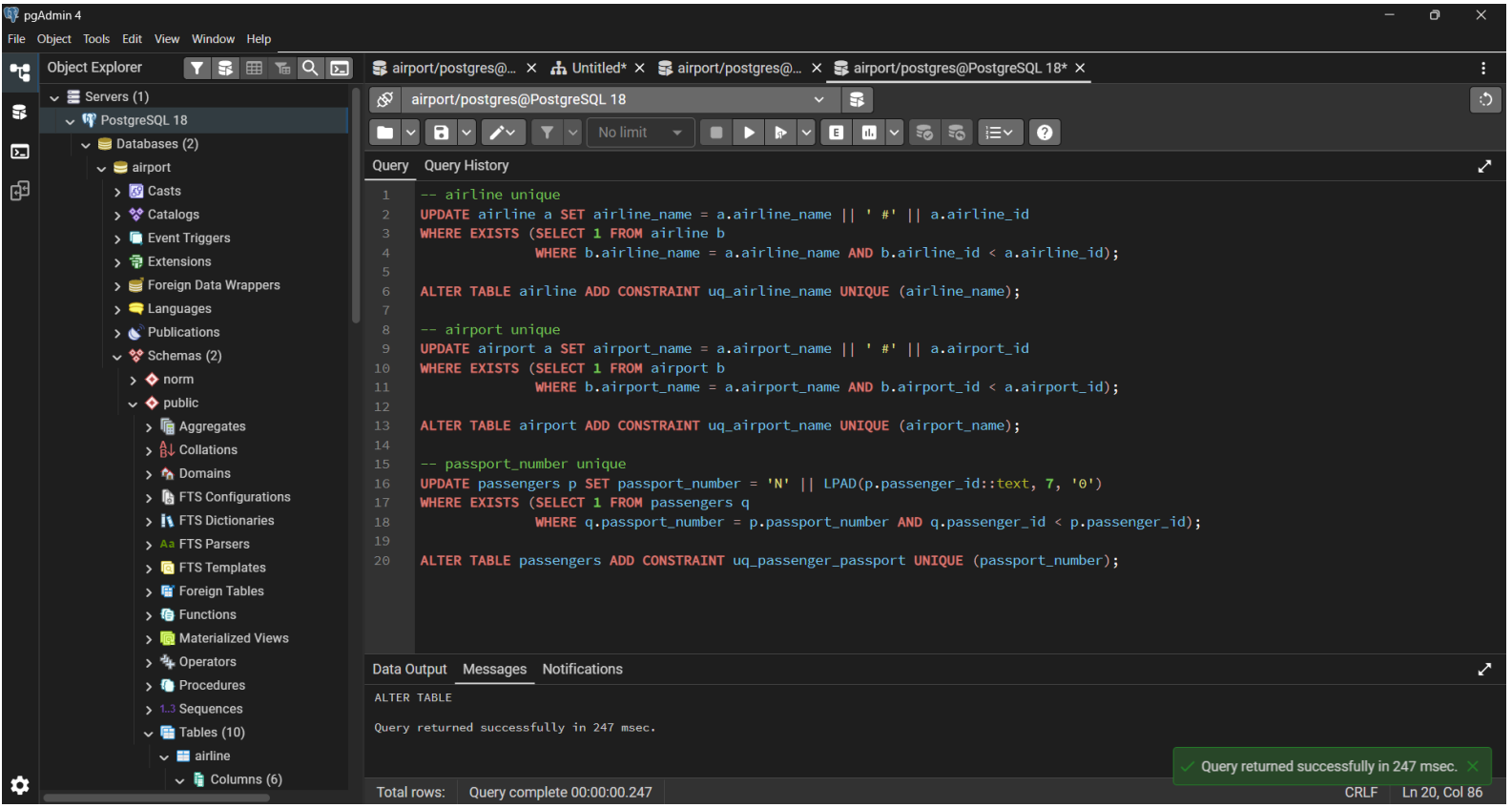
Task 3

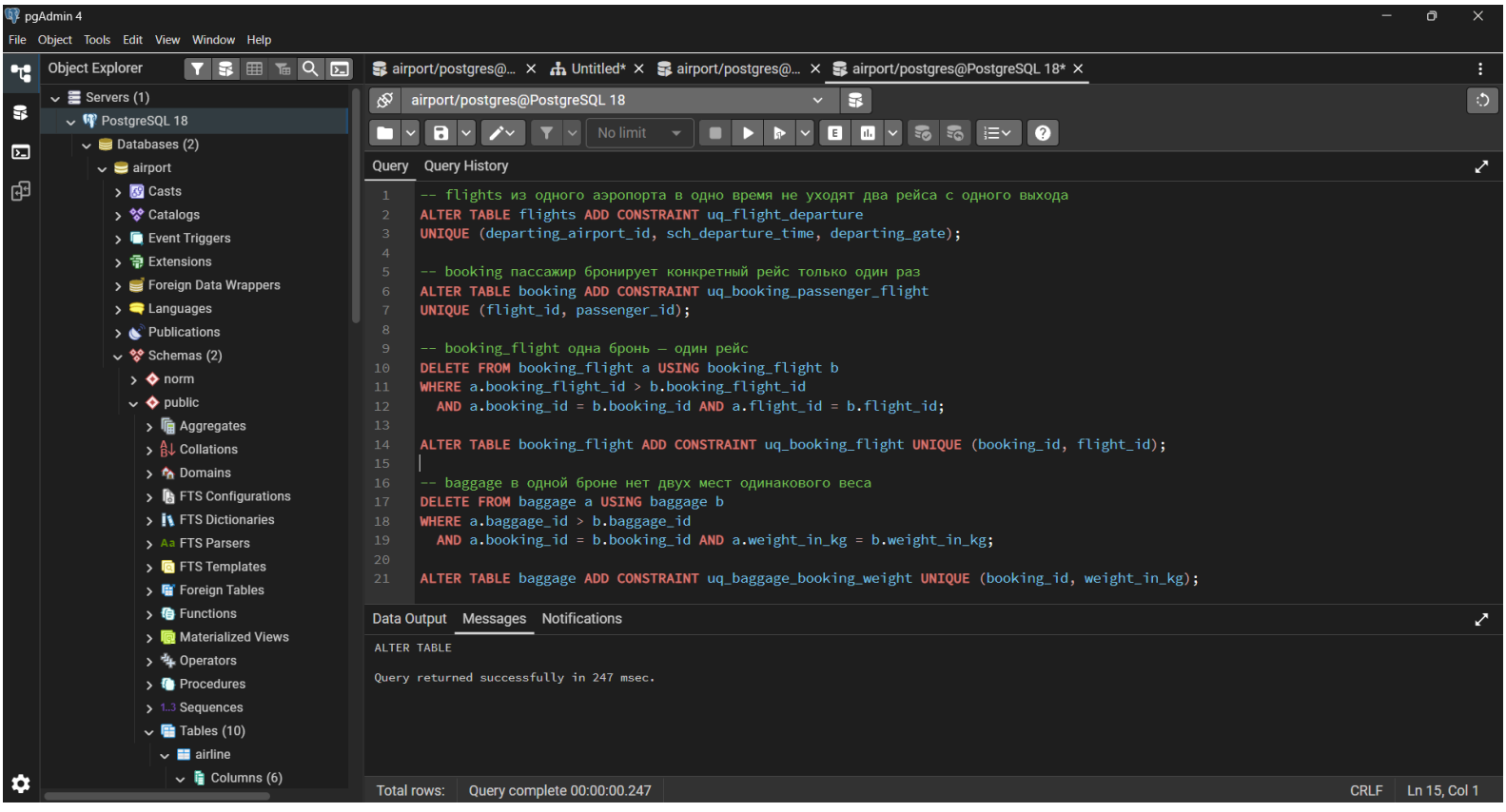

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airport/postgres@PostgreSQL 18
No limit
Query Query History
UPDATE baggage SET weight_in_kg = 23.00, updated_at = now() WHERE weight_in_kg > 23;
UPDATE baggage SET weight_in_kg = 1.00, updated_at = now() WHERE weight_in_kg < 1;
ALTER TABLE baggageADD CONSTRAINT chk_baggage_weight_range CHECK (weight_in_kg BETWEEN 1 AND 23);
SELECT baggage_id, weight_in_kg FROM baggage ORDER BY weight_in_kg DESC;
Data Output Messages Notifications
SQL
Showing rows: 1 to 8
Page No: 1 of 1
baggage_id [PK] integer
weight_in_kg numeric (4,2)
1 2 23.00
2 5 23.00
3 7 23.00
4 10 21.70
5 1 15.11
6 4 14.99
7 6 8.08
8 8 6.03
Total rows: 8
Query complete 00:00:00.881
CRLF
Ln 4, Col 60

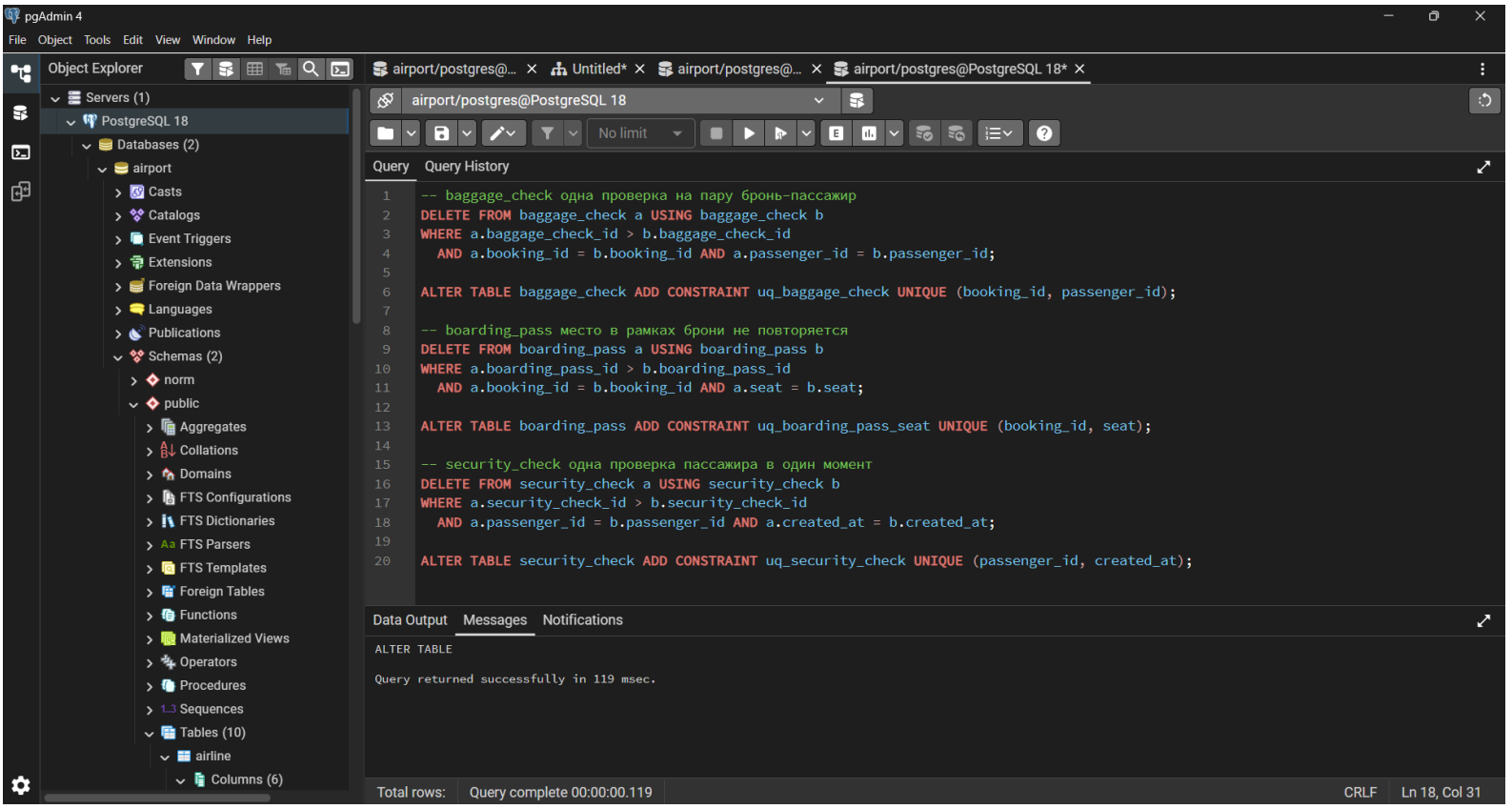
Task 4

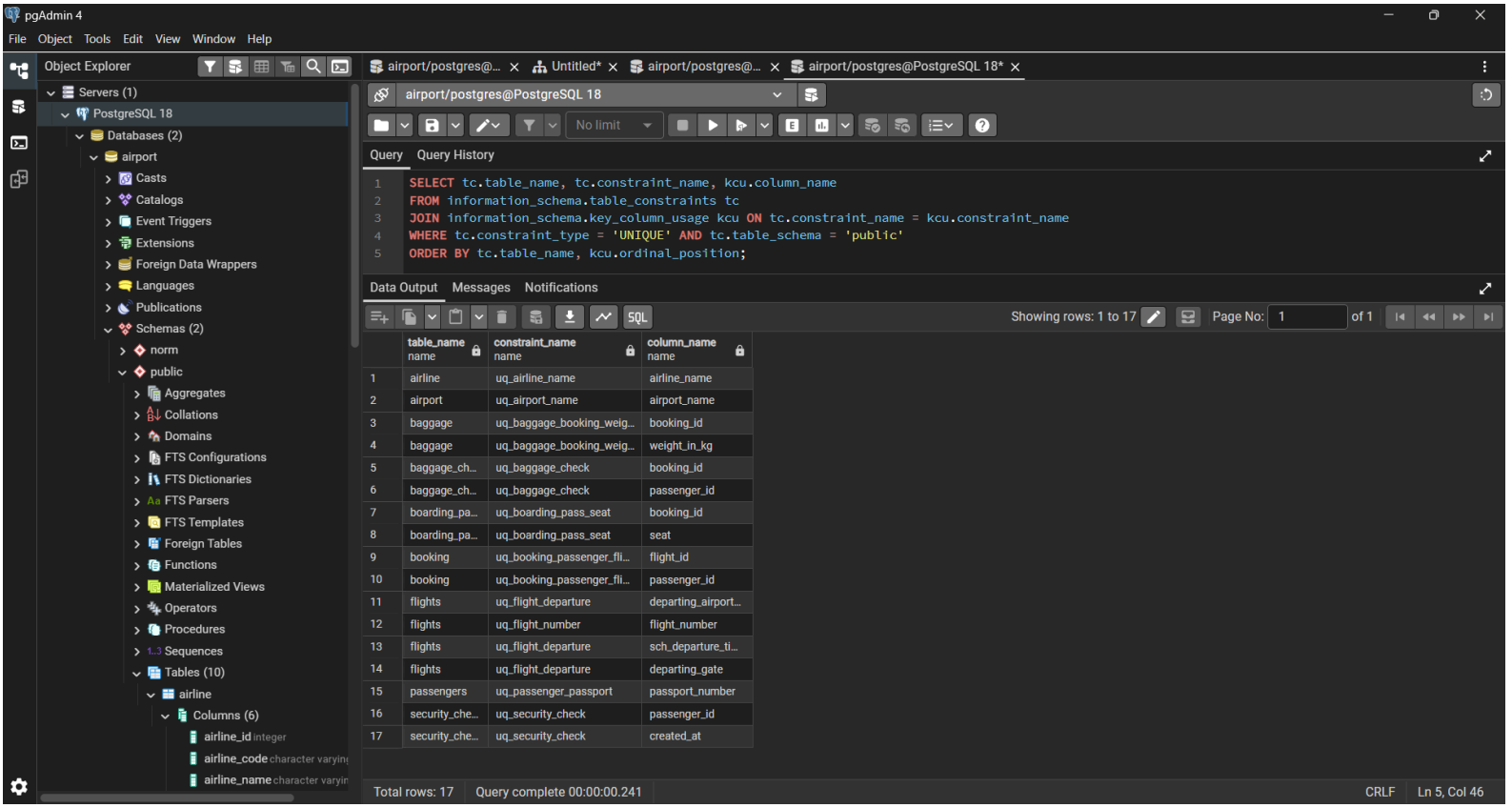

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Columns (6)
airport/postgres@... Untitled* airport/postgres@... airport/postgres@PostgreSQL 18*
airport/postgres@PostgreSQL 18
No limit
Query Query History
ALTER TABLE airport
ADD CONSTRAINT chk_airport_name_length
CHECK (LENGTH(airport_name) >= 10);
SELECT airport_id, airport_name, LENGTH(airport_name) AS name_length FROM airport ORDER BY name_length;
Data Output Messages Notifications
SQL
Showing rows: 1 to 18
Page No: 1 of 1
airport_id [PK] integer
airport_name character varying (50)
name_length integer
1 11 Chopin Airport 14
2 5 Narita Airport 14
3 19 Shymkent Airport 16
4 4 Heathrow Airport 16
5 3 Pavlodar Airport 16
6 14 Istanbul Airport 16
7 10 Schiphol Airport 16
8 20 Sary-Arka Airport 17
9 12 Sheremetyevo Airport 20
10 17 Helsinki-Vantaa Airport 23
Total rows: 18
Query complete 00:00:00.217
CRLF
Ln 5, Col 104

Task 5

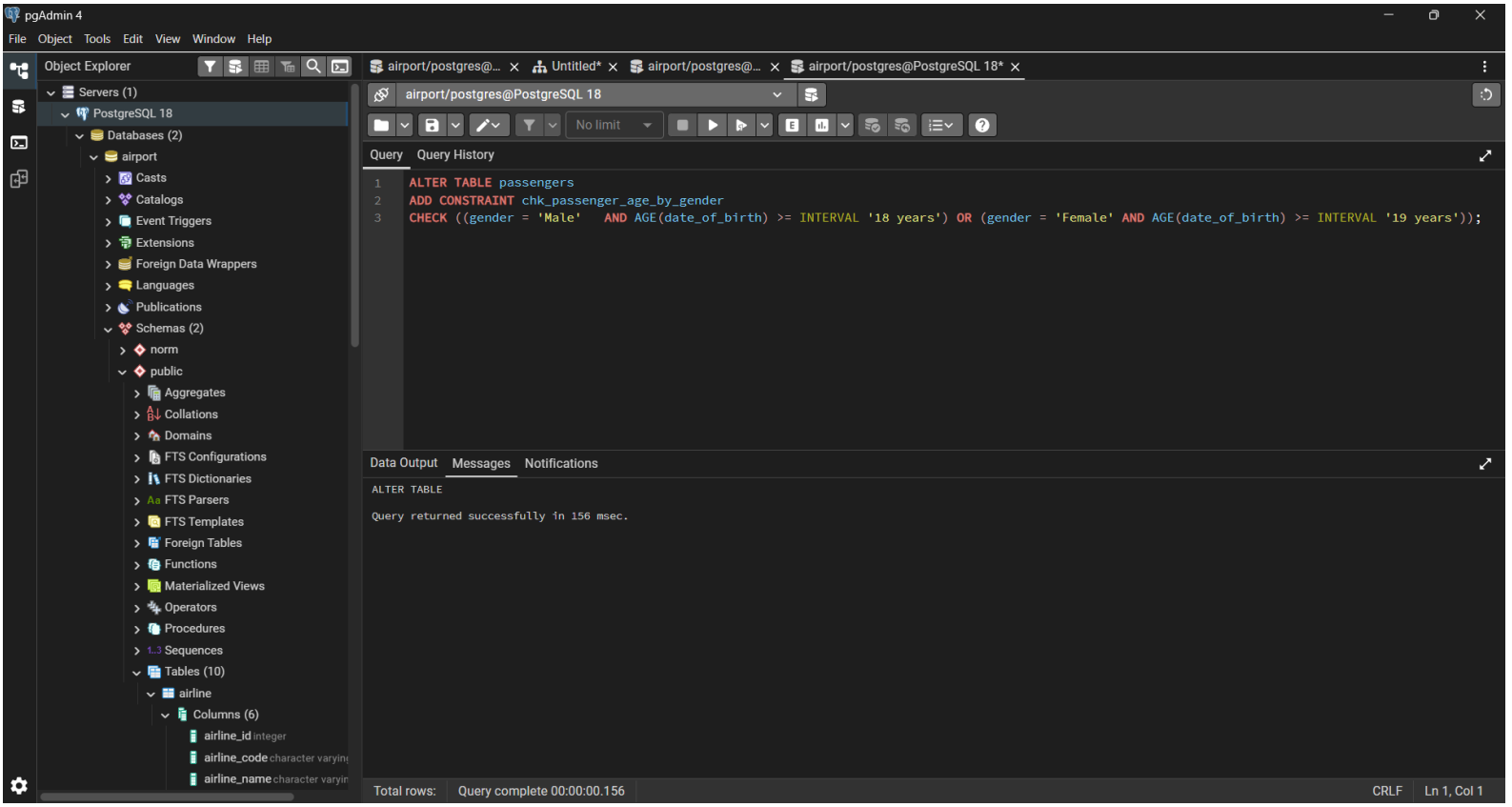

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Columns (6)
airport/postgres@... × Untitled* × airport/postgres@... × airport/postgres@PostgreSQL 18* ×
airport/postgres@PostgreSQL 18
No limit
Query Query History
1 -- airline unique
2 UPDATE airline a SET airline_name = a.airline_name || ' #' || a.airline_id
3 WHERE EXISTS (SELECT 1 FROM airline b
4 WHERE b.airline_name = a.airline_name AND b.airline_id < a.airline_id);
5
6 ALTER TABLE airline ADD CONSTRAINT uq_airline_name UNIQUE (airline_name);
7
8 -- airport unique
9 UPDATE airport a SET airport_name = a.airport_name || ' #' || a.airport_id
10 WHERE EXISTS (SELECT 1 FROM airport b
11 WHERE b.airport_name = a.airport_name AND b.airport_id < a.airport_id);
12
13 ALTER TABLE airport ADD CONSTRAINT uq_airport_name UNIQUE (airport_name);
14
15 -- passport_number unique
16 UPDATE passengers p SET passport_number = 'N' || LPAD(p.passenger_id::text, 7, '0')
17 WHERE EXISTS (SELECT 1 FROM passengers q
18 WHERE q.passport_number = p.passport_number AND q.passenger_id < p.passenger_id);
19
20 ALTER TABLE passengers ADD CONSTRAINT uq_passenger_passport UNIQUE (passport_number);
Data Output Messages Notifications
ALTER TABLE
Query returned successfully in 247 msec.
Query returned successfully in 247 msec.
Total rows: Query complete 00:00:00.247
CRLF Ln 20, Col 86


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airline
Columns (6)
airport/postgres@... × Untitled* × airport/postgres@... × airport/postgres@PostgreSQL 18* ×
airport/postgres@PostgreSQL 18
No limit
Query Query History
1 -- flights из одного аэропорта в одно время не уходят два рейса с одного выхода
2 ALTER TABLE flights ADD CONSTRAINT uq_flight_departure
3 UNIQUE (departing_airport_id, sch_departure_time, departing_gate);
4
5 -- booking пассажир бронирует конкретный рейс только один раз
6 ALTER TABLE booking ADD CONSTRAINT uq_booking_passenger_flight
7 UNIQUE (flight_id, passenger_id);
8
9 -- booking_flight одна бронь — один рейс
10 DELETE FROM booking_flight a USING booking_flight b
11 WHERE a.booking_flight_id > b.booking_flight_id
12 AND a.booking_id = b.booking_id AND a.flight_id = b.flight_id;
13
14 ALTER TABLE booking_flight ADD CONSTRAINT uq_booking_flight UNIQUE (booking_id, flight_id);
15
16 -- baggage в одной броне нет двух мест одинакового веса
17 DELETE FROM baggage a USING baggage b
18 WHERE a.baggage_id > b.baggage_id
19 AND a.booking_id = b.booking_id AND a.weight_in_kg = b.weight_in_kg;
20
21 ALTER TABLE baggage ADD CONSTRAINT uq_baggage_booking_weight UNIQUE (booking_id, weight_in_kg);
Data Output Messages Notifications
ALTER TABLE
Query returned successfully in 247 msec.
Total rows: Query complete 00:00:00.247
CRLF Ln 15, Col 1

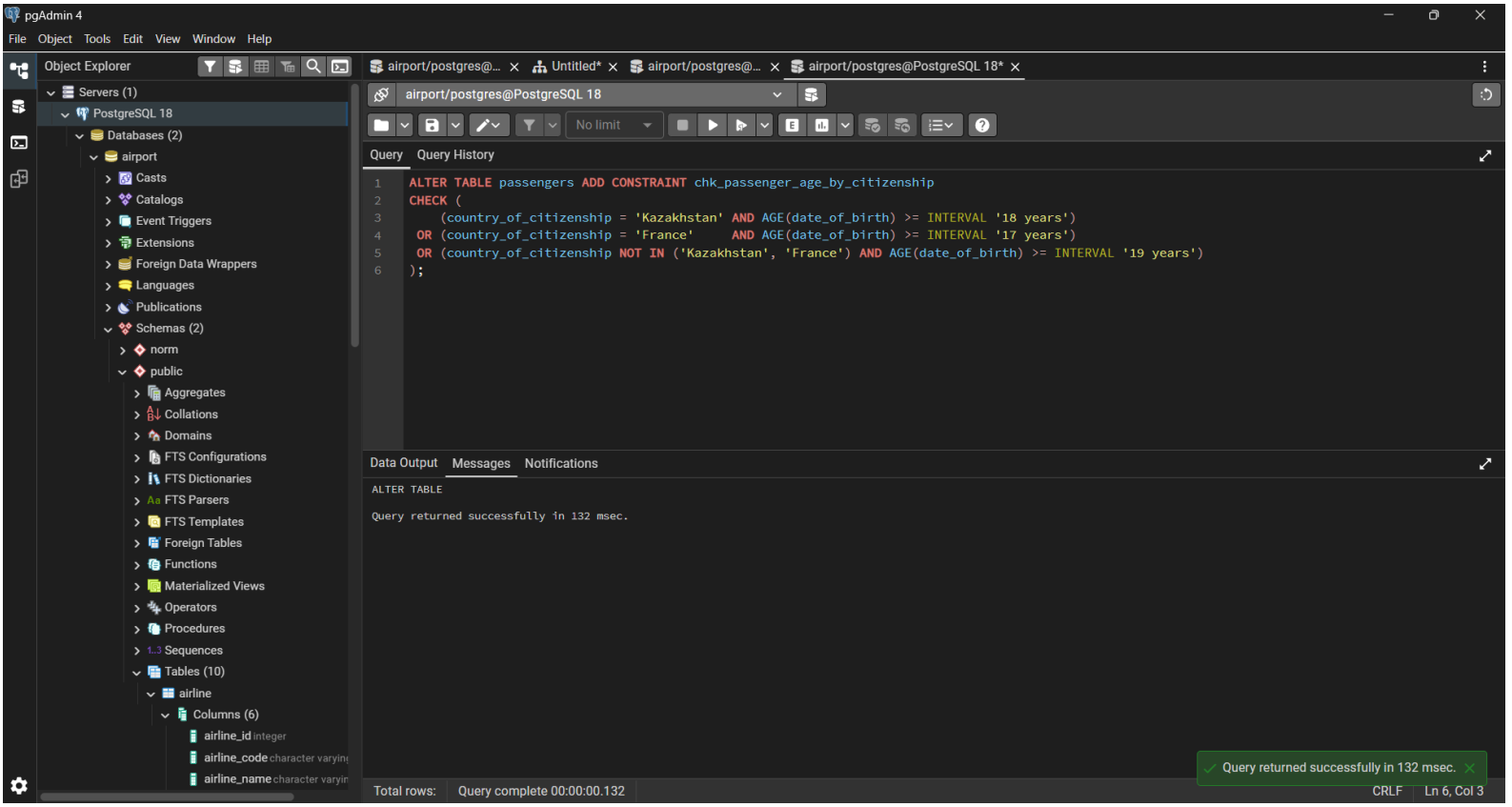

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Columns (6)
airport/postgres@... × Untitled* × airport/postgres@... × airport/postgres@PostgreSQL 18* ×
airport/postgres@PostgreSQL 18
No limit
Query Query History
1 -- baggage_check одна проверка на пару бронь-пассажир
2 DELETE FROM baggage_check a USING baggage_check b
3 WHERE a.baggage_check_id > b.baggage_check_id
4 AND a.booking_id = b.booking_id AND a.passenger_id = b.passenger_id;
5
6 ALTER TABLE baggage_check ADD CONSTRAINT uq_baggage_check UNIQUE (booking_id, passenger_id);
7
8 -- boarding_pass место в рамках брони не повторяется
9 DELETE FROM boarding_pass a USING boarding_pass b
10 WHERE a.boarding_pass_id > b.boarding_pass_id
11 AND a.booking_id = b.booking_id AND a.seat = b.seat;
12
13 ALTER TABLE boarding_pass ADD CONSTRAINT uq_boarding_pass_seat UNIQUE (booking_id, seat);
14
15 -- security_check одна проверка пассажира в один момент
16 DELETE FROM security_check a USING security_check b
17 WHERE a.security_check_id > b.security_check_id
18 AND a.passenger_id = b.passenger_id AND a.created_at = b.created_at;
19
20 ALTER TABLE security_check ADD CONSTRAINT uq_security_check UNIQUE (passenger_id, created_at);
Data Output Messages Notifications
ALTER TABLE
Query returned successfully in 119 msec.
Total rows: Query complete 00:00:00.119 CRLF Ln 18, Col 31


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Tables (10)
airline
Columns (6)
airline_id integer
airline_code character varying
airline_name character varying
airport/postgres@... × Untitled* × airport/postgres@... × airport/postgres@PostgreSQL 18* ×
airport/postgres@PostgreSQL 18
No limit
Query Query History
1 SELECT tc.table_name, tc.constraint_name, kcu.column_name
2 FROM information_schema.table_constraints tc
3 JOIN information_schema.key_column_usage kcu ON tc.constraint_name = kcu.constraint_name
4 WHERE tc.constraint_type = 'UNIQUE' AND tc.table_schema = 'public'
5 ORDER BY tc.table_name, kcu.ordinal_position;
Data Output Messages Notifications
SQL
Showing rows: 1 to 17 Page No: 1 of 1
table_name name
constraint_name name
column_name name
1 airline uq_airline_name airline_name
2 airport uq_airport_name airport_name
3 baggage uq_baggage_booking_weig... booking_id
4 baggage uq_baggage_booking_weig... weight_in_kg
5 baggage_ch... uq_baggage_check booking_id
6 baggage_ch... uq_baggage_check passenger_id
7 boarding_pa... uq_boarding_pass_seat booking_id
8 boarding_pa... uq_boarding_pass_seat seat
9 booking uq_booking_passenger_fli... flight_id
10 booking uq_booking_passenger_fli... passenger_id
11 flights uq_flight_departure departing_airport...
12 flights uq_flight_number flight_number
13 flights uq_flight_departure sch_departure_ti...
14 flights uq_flight_departure departing_gate
15 passengers uq_passenger_passport passport_number
16 security_che... uq_security_check passenger_id
17 security_che... uq_security_check created_at
Total rows: 17 Query complete 00:00:00.241 CRLF Ln 5, Col 46

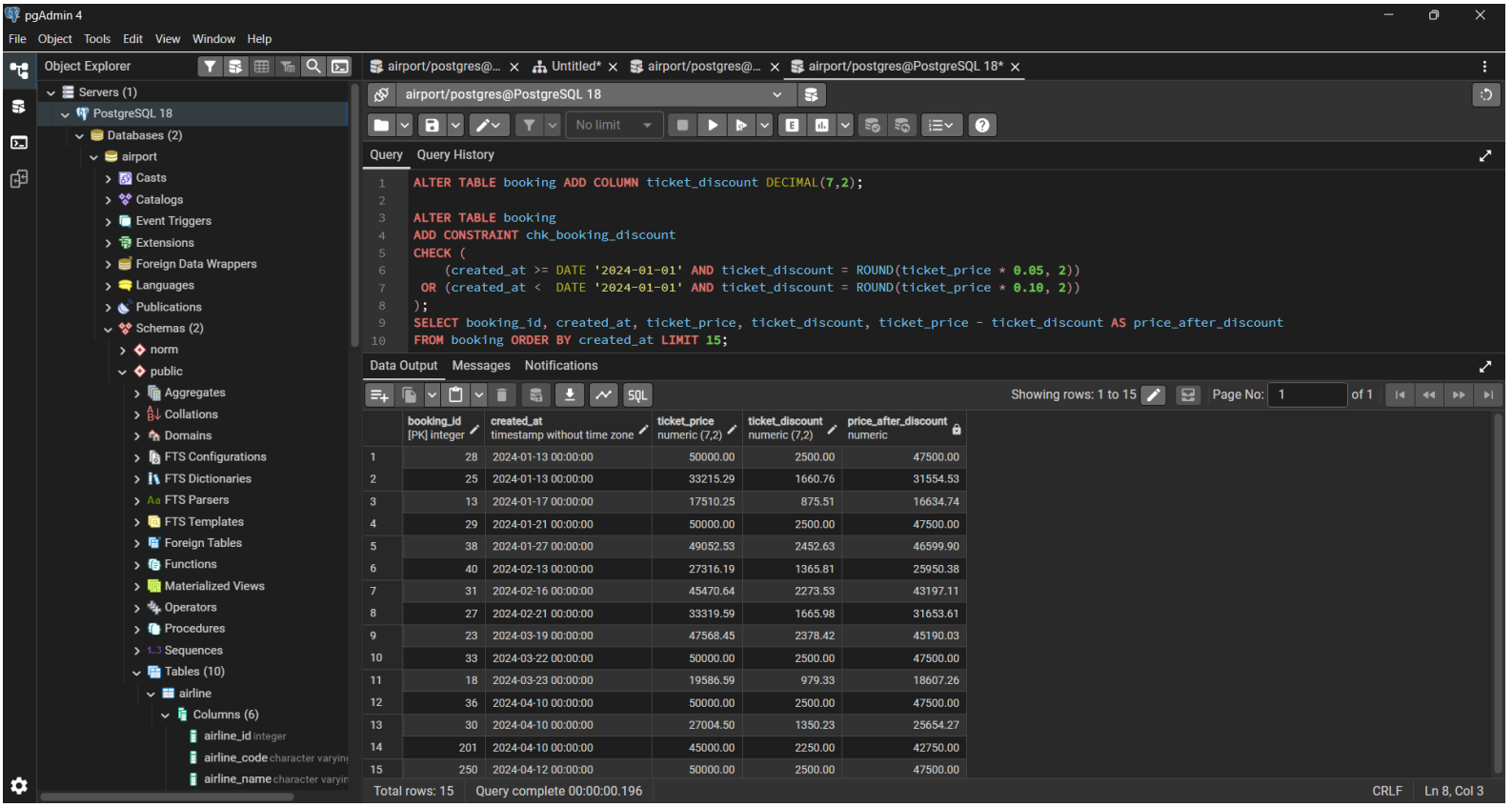
Task 6


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Tables (10)
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Columns (6)
airline_id integer
airline_code character varying
airline_name character varyin
airport/postgres@... Untitled* airport/postgres@... airport/postgres@PostgreSQL 18*
airport/postgres@PostgreSQL 18
No limit
Query Query History
1 ALTER TABLE passengers
2 ADD CONSTRAINT chk_passenger_age_by_gender
3 CHECK ((gender = 'Male' AND AGE(date_of_birth) >= INTERVAL '18 years') OR (gender = 'Female' AND AGE(date_of_birth) >= INTERVAL '19 years'));
Data Output Messages Notifications
ALTER TABLE
Query returned successfully in 156 msec.
Total rows: Query complete 00:00:00.156
CRLF Ln 1, Col 1

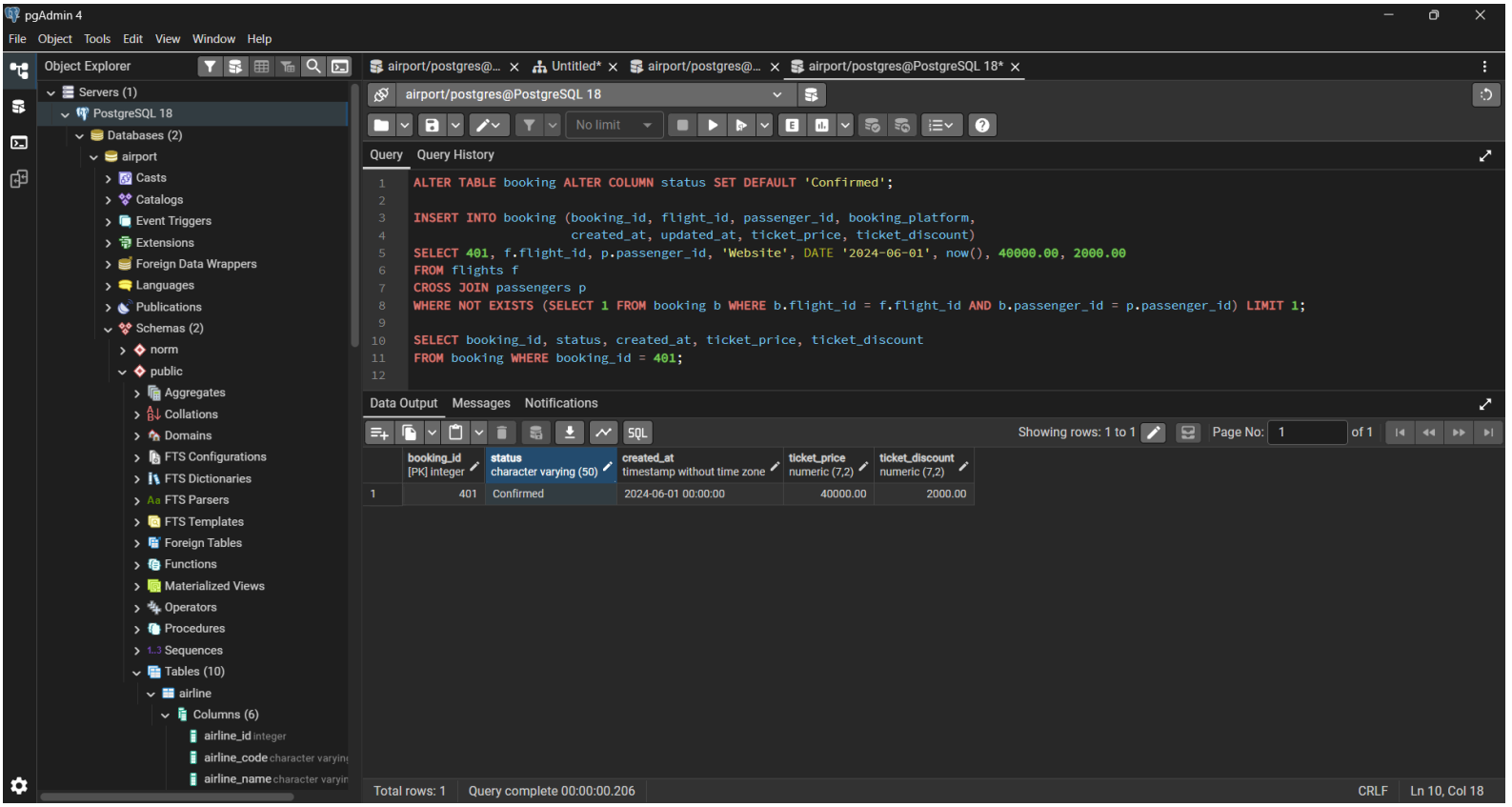
Task 7


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Columns (6)
airline_id integer
airline_code character varying
airline_name character varyin
airport/postgres@... Untitled* airport/postgres@... airport/postgres@PostgreSQL 18*
airport/postgres@PostgreSQL 18
No limit
Query Query History
1 ALTER TABLE passengers ADD CONSTRAINT chk_passenger_age_by_citizenship
2 CHECK (
3 (country_of_citizenship = 'Kazakhstan' AND AGE(date_of_birth) >= INTERVAL '18 years')
4 OR (country_of_citizenship = 'France' AND AGE(date_of_birth) >= INTERVAL '17 years')
5 OR (country_of_citizenship NOT IN ('Kazakhstan', 'France') AND AGE(date_of_birth) >= INTERVAL '19 years')
6);
Data Output Messages Notifications
ALTER TABLE
Query returned successfully in 132 msec.
Query returned successfully in 132 msec.
Total rows: Query complete 00:00:00.132
CRLF Ln 6, Col 3

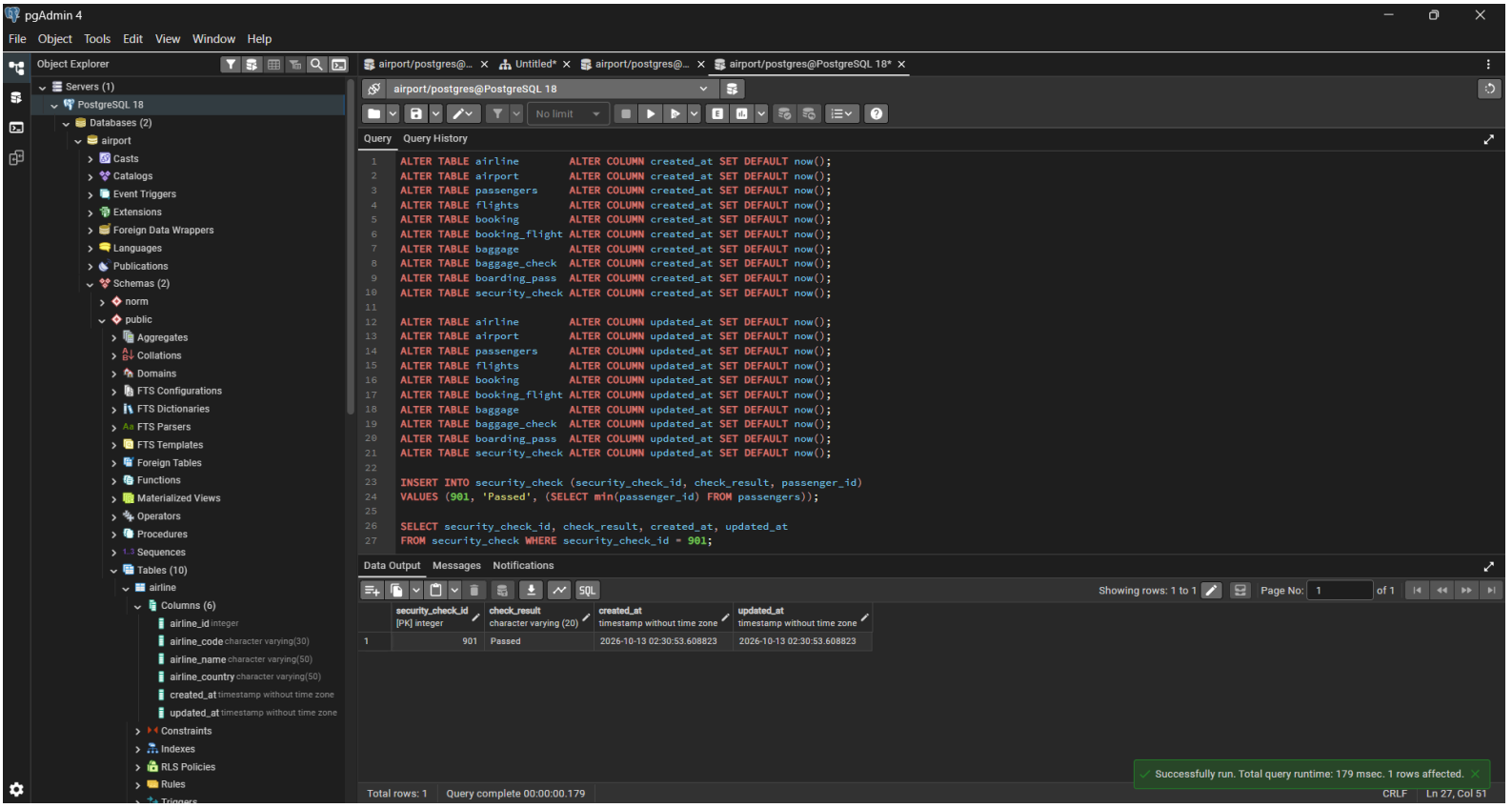
Task 8

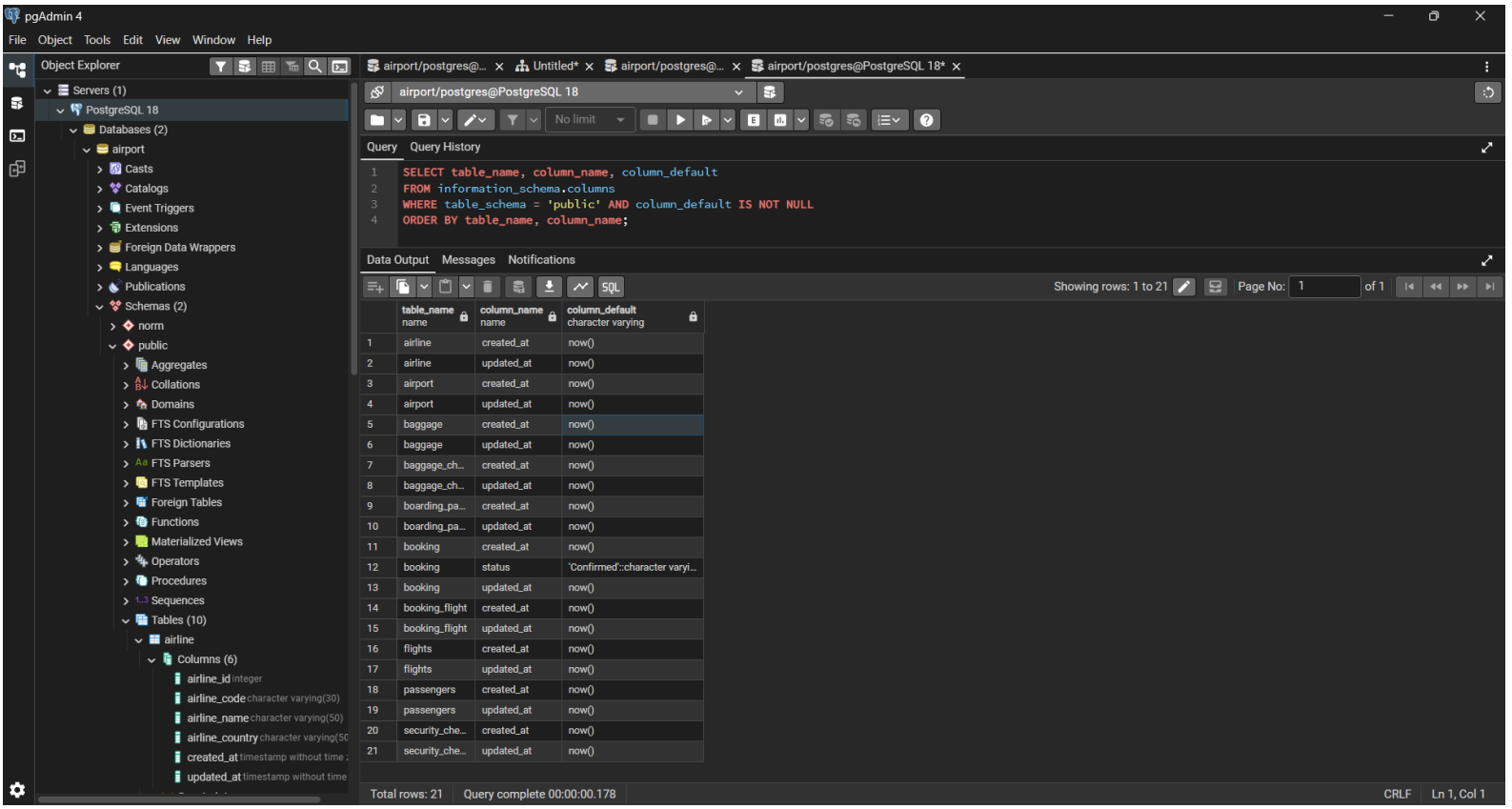

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airline_id integer
airline_code character varying
airline_name character varyin
airport/postgres@... Untitled* airport/postgres@... airport/postgres@PostgreSQL 18*
airport/postgres@PostgreSQL 18
No limit
Query Query History
1 ALTER TABLE booking ADD COLUMN ticket_discount DECIMAL(7,2);
2
3 ALTER TABLE booking
4 ADD CONSTRAINT chk_booking_discount
5 CHECK (
6 (created_at >= DATE '2024-01-01' AND ticket_discount = ROUND(ticket_price * 0.05, 2))
7 OR (created_at < DATE '2024-01-01' AND ticket_discount = ROUND(ticket_price * 0.10, 2))
8);
9 SELECT booking_id, created_at, ticket_price, ticket_discount, ticket_price - ticket_discount AS price_after_discount
10 FROM booking ORDER BY created_at LIMIT 15;
Data Output Messages Notifications
SQL
Showing rows: 1 to 15 Page No: 1 of 1
booking_id [PK] integer | created_at timestamp without time zone | ticket_price numeric (7,2) | ticket_discount numeric (7,2) | price_after_discount numeric
1 28 2024-01-13 00:00:00 50000.00 2500.00 47500.00
2 25 2024-01-13 00:00:00 33215.29 1660.76 31554.53
3 13 2024-01-17 00:00:00 17510.25 875.51 16634.74
4 29 2024-01-21 00:00:00 50000.00 2500.00 47500.00
5 38 2024-01-27 00:00:00 49052.53 2452.63 46599.90
6 40 2024-02-13 00:00:00 27316.19 1365.81 25950.38
7 31 2024-02-16 00:00:00 45470.64 2273.53 43197.11
8 27 2024-02-21 00:00:00 33319.59 1665.98 31653.61
9 23 2024-03-19 00:00:00 47568.45 2378.42 45190.03
10 33 2024-03-22 00:00:00 50000.00 2500.00 47500.00
11 18 2024-03-23 00:00:00 19586.59 979.33 18607.26
12 36 2024-04-10 00:00:00 50000.00 2500.00 47500.00
13 30 2024-04-10 00:00:00 27004.50 1350.23 25654.27
14 201 2024-04-10 00:00:00 45000.00 2250.00 42750.00
15 250 2024-04-12 00:00:00 50000.00 2500.00 47500.00
Total rows: 15 Query complete 00:00:00.196
CRLF Ln 8, Col 3

Task 9


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airline_code character varying
airline_name character varyin
airport/postgres@... Untitled* airport/postgres@... airport/postgres@PostgreSQL 18*
airport/postgres@PostgreSQL 18
No limit
Query Query History
1 ALTER TABLE booking ALTER COLUMN status SET DEFAULT 'Confirmed';
2
3 INSERT INTO booking (booking_id, flight_id, passenger_id, booking_platform,
4 created_at, updated_at, ticket_price, ticket_discount)
5 SELECT 401, f.flight_id, p.passenger_id, 'Website', DATE '2024-06-01', now(), 40000.00, 2000.00
6 FROM flights f
7 CROSS JOIN passengers p
8 WHERE NOT EXISTS (SELECT 1 FROM booking b WHERE b.flight_id = f.flight_id AND b.passenger_id = p.passenger_id) LIMIT 1;
9
10 SELECT booking_id, status, created_at, ticket_price, ticket_discount
11 FROM booking WHERE booking_id = 401;
12
Data Output Messages Notifications
SQL
Showing rows: 1 to 1 Page No: 1 of 1
booking_id [PK] integer | status character varying (50) | created_at timestamp without time zone | ticket_price numeric (7,2) | ticket_discount numeric (7,2)
1 401 Confirmed 2024-06-01 00:00:00 40000.00 2000.00
Total rows: 1 Query complete 00:00:00.206
CRLF Ln 10, Col 18

Task 10


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Columns (6)
airline_id integer
airline_code character varying(30)
airline_name character varying(50)
airline_country character varying(50)
created_at timestamp without time zone
updated_at timestamp without time zone
airport/postgres@PostgreSQL 18
Query Query History
ALTER TABLE airline ALTER COLUMN created_at SET DEFAULT now();
ALTER TABLE airport ALTER COLUMN created_at SET DEFAULT now();
ALTER TABLE passengers ALTER COLUMN created_at SET DEFAULT now();
ALTER TABLE flights ALTER COLUMN created_at SET DEFAULT now();
ALTER TABLE booking ALTER COLUMN created_at SET DEFAULT now();
ALTER TABLE booking_flight ALTER COLUMN created_at SET DEFAULT now();
ALTER TABLE baggage ALTER COLUMN created_at SET DEFAULT now();
ALTER TABLE baggage_check ALTER COLUMN created_at SET DEFAULT now();
ALTER TABLE boarding_pass ALTER COLUMN created_at SET DEFAULT now();
ALTER TABLE security_check ALTER COLUMN created_at SET DEFAULT now();
ALTER TABLE airline ALTER COLUMN updated_at SET DEFAULT now();
ALTER TABLE airport ALTER COLUMN updated_at SET DEFAULT now();
ALTER TABLE passengers ALTER COLUMN updated_at SET DEFAULT now();
ALTER TABLE flights ALTER COLUMN updated_at SET DEFAULT now();
ALTER TABLE booking ALTER COLUMN updated_at SET DEFAULT now();
ALTER TABLE booking_flight ALTER COLUMN updated_at SET DEFAULT now();
ALTER TABLE baggage ALTER COLUMN updated_at SET DEFAULT now();
ALTER TABLE baggage_check ALTER COLUMN updated_at SET DEFAULT now();
ALTER TABLE boarding_pass ALTER COLUMN updated_at SET DEFAULT now();
ALTER TABLE security_check ALTER COLUMN updated_at SET DEFAULT now();
INSERT INTO security_check (security_check_id, check_result, passenger_id)
VALUES (901, 'Passed', (SELECT min(passenger_id) FROM passengers));
SELECT security_check_id, check_result, created_at, updated_at
FROM security_check WHERE security_check_id = 901;
Data Output Messages Notifications
security_check_id [PK] integer
check_result character varying (20)
created_at timestamp without time zone
updated_at timestamp without time zone
1 901 Passed 2026-10-13 02:30:53.608823 2026-10-13 02:30:53.608823
Successfully run. Total query runtime: 179 msec. 1 rows affected.
Total rows: 1 Query complete 00:00:00.179
CRLF Ln 27, Col 51


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Query Query History
SELECT table_name, column_name, column_default
FROM information_schema.columns
WHERE table_schema = 'public' AND column_default IS NOT NULL
ORDER BY table_name, column_name;
Data Output Messages Notifications
table_name name
column_name name
column_default character varying
1 airline created_at now()
2 airline updated_at now()
3 airport created_at now()
4 airport updated_at now()
5 baggage created_at now()
6 baggage updated_at now()
7 baggage_ch... created_at now()
8 baggage_ch... updated_at now()
9 boarding_pa... created_at now()
10 boarding_pa... updated_at now()
11 booking created_at now()
12 booking status 'Confirmed'::character varyi...
13 booking updated_at now()
14 booking_flight created_at now()
15 booking_flight updated_at now()
16 flights created_at now()
17 flights updated_at now()
18 passengers created_at now()
19 passengers updated_at now()
20 security_che... created_at now()
21 security_che... updated_at now()
Total rows: 21 Query complete 00:00:00.178
CRLF Ln 1, Col 1